

Retrieval – Augmented Generation (RAG)

Student Details

Name: Georgios Kitsakis

Email: geo.kitsakis@aueb.gr

Dataset: Climate Change Document

Colab Implementation

Google Colab Link:

https://drive.google.com/file/d/13Ch9YU3Kv-B06q_fde-6ihtcj3wrWlQl/view?usp=sharing

This assignment implements an end-to-end RAG pipeline for answering climate change questions. The notebook includes:

1. Document Processing: Extracted and chunked 10 climate change text files into 211 semantic chunks (400 chars, 50 overlap).
2. Embedding & Vector DB: Generated embeddings using all-MiniLM-L6-v2 model and stored in FAISS vector database.
3. Rag Function: Implemented ask(q) function that retrieves top-k relevant chunks and generates answers using TinyLlama 1.1B.
4. Evaluation: Validated against 44 reference Q&A pairs with high accuracy and context-grounded answers.